

Actual vs Budget Analysis

SQL + Power BI

**Presented by
Busayo Omoniyi**

Management needs visibility into how effectively the company is managing its allocated budget and where unfavorable spending patterns are occurring.

This project analyzes three years of financial transaction data to compare budgeted amounts against actual spending and identify the departments, expense categories, periods and regions contributing to budget overruns.

Objective

Transform raw sales data into meaningful business insights through data cleaning, SQL analysis, interactive dashboards, and data storytelling.

Project Overview

Reporting Period:

January 2021 –
December 2023

Records Analyzed:

10,000

Tools:

SQL, Power BI, Excel

ROLE:

Data Analyst



This analysis was conducted to answer the above business questions

Data Preparation & Quality Assessment

The original dataset contained 10,010 records and 8 columns covering transactions from January 2021 to December 2023.

Initial data quality checks identified:

- 10 exact duplicate records
- 8 missing values
- Missing values across Region, Transaction ID, Category and Department

Cleaning Approach

Using SQL, exact duplicate records were removed and missing categorical values were standardized as "Unknown" rather than being arbitrarily assigned to a business category.

The cleaned dataset contained 10,000 records and was then used for analysis and Power BI reporting.

Analytical Approach

The analysis followed an end-to-end workflow:

Raw Data → SQL Data Quality Assessment → Cleaning → Business Questions → SQL Analysis → Validation → Power BI Modeling → Dashboard → Insights → Recommendations

Budget variance was calculated as: Actual Spending – Budget

Therefore, Positive variance = Overspending

Negative variance = Spending below budget

This convention was used consistently throughout the analysis.

Key Findings

1. The company overspent by ₦94.8M

Across the three-year period;

Total Budget	-	₦795.47M
Actual Spending	-	₦890.27M
Total Overspend	-	₦94.80M
Variance	-	11.92%

Actual spending exceeded budget consistently throughout the reporting period.

Executive Dashboard

Executive Dashboard

Time & Budget Perf.

Department & Expense...

Regional Performance &...

Glossary

₦ 890.27M

Total Spend

₦ 795.47M

Total Budget

₦ 94.80M

Total Overspend

11.9%

Variance %

2022

Worst Performing Year

Finance

HR

IT

Marketing

Operations

Sales

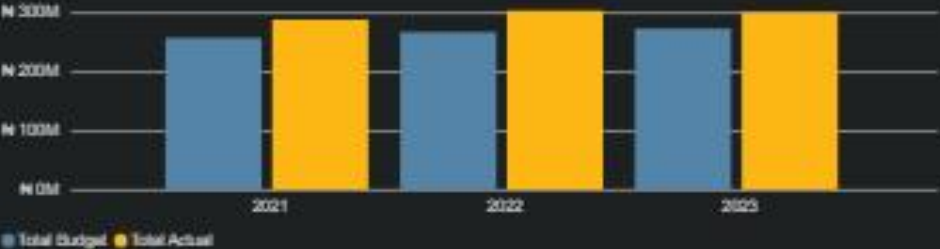
Unknown

Actual Spending vs Budget

Monthly performance, Jan 2021–Dec 2023



Budget Vs Actual



Budget Variance Yearly Pattern



2. 2022 recorded the worst annual budget performance

Annual overspending was:

Year	Overspend
2021	₦30.83M
2022	₦36.43M
2023	₦27.53M

2022 recorded the highest overspend at ₦36.43M, representing approximately 13.68% of its annual budget.

Although performance improved in 2023, the company remained over budget.



Time & Budget Performance

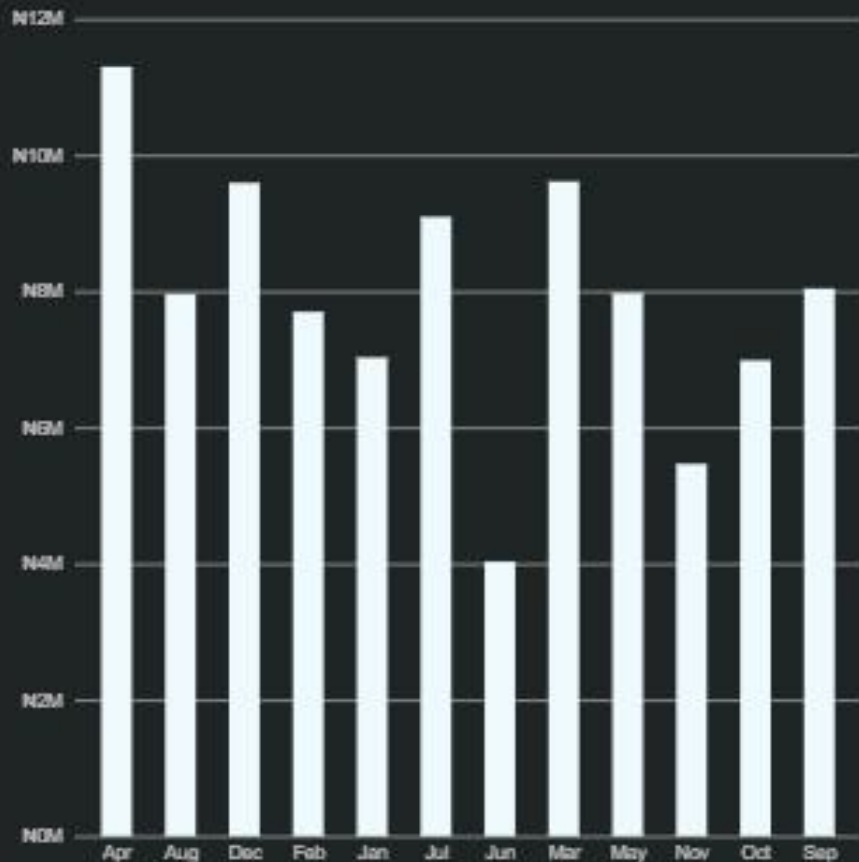
₦ 36.43M

Largest Annual Overspend

2022

Worst Performing Year

Monthly Overspend



Year	Total Actual	Total Budget	BV
2021	₦ 287,511,086	₦ 256,681,020	₦ 30,830,066
2022			
January	₦ 26,727,517	₦ 23,546,292	₦ 3,181,225
February	₦ 21,186,138	₦ 19,296,156	₦ 1,889,982
March	₦ 24,891,811	₦ 20,905,031	₦ 3,986,780
April	₦ 24,055,215	₦ 20,561,645	₦ 3,493,570
May	₦ 26,605,506	₦ 22,751,826	₦ 3,853,680
June	₦ 24,875,581	₦ 22,687,896	₦ 2,187,685
July	₦ 26,115,690	₦ 23,703,274	₦ 2,412,416
August	₦ 26,201,877	₦ 22,779,028	₦ 3,422,849
September	₦ 23,615,228	₦ 21,101,758	₦ 2,513,470
October	₦ 26,499,256	₦ 24,781,493	₦ 1,717,763

23.50%
Top 5 Overspend Contribution

Top 5 Month by Overspend



3. Marketing was the largest overspending department

Marketing recorded the highest cumulative departmental overspend:

₦19.88M | 15.06%

Other departments also recorded unfavorable variances, indicating that overspending was not isolated to one department.

Finance recorded the smallest departmental overspend at ₦11.14M (8.38%).



Department & Expense Drivers

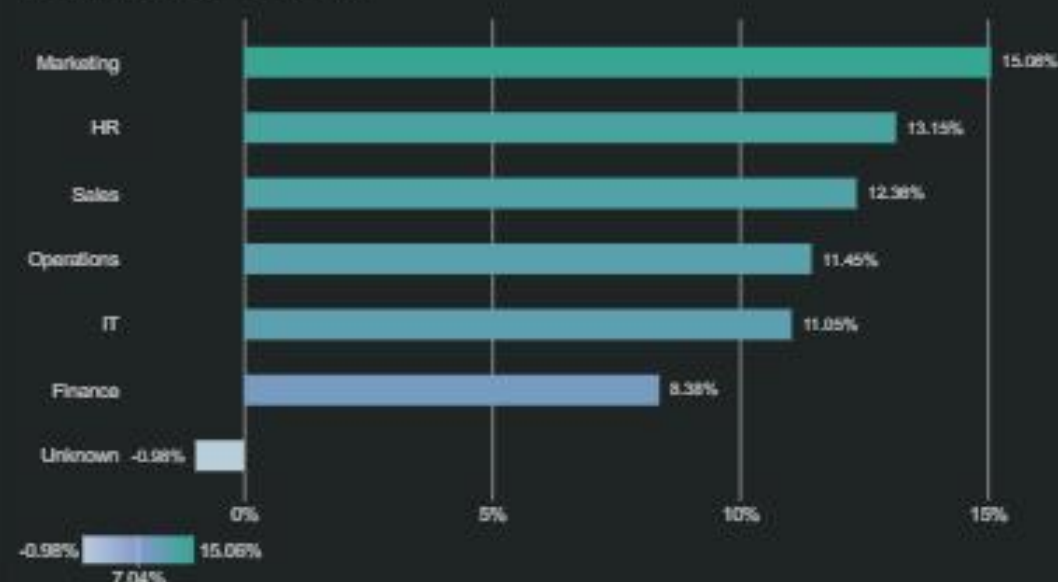
Actual spending & Budget Overview by Department

Department	Total Actual	Total Budget	Variance	Variance %
Marketing	¥ 151,816,098	¥ 131,939,479	¥ 19,876,619	0.15
HR	¥ 155,439,763	¥ 137,371,163	¥ 18,068,600	0.13
Sales	¥ 151,344,999	¥ 134,702,488	¥ 16,642,511	0.12
Operations	¥ 147,803,987	¥ 132,622,213	¥ 15,181,774	0.11
IT	¥ 139,663,026	¥ 125,771,001	¥ 13,892,025	0.11
Finance	¥ 144,084,529	¥ 132,947,149	¥ 11,137,380	0.08
Unknown	¥ 114,627	¥ 115,763	-¥1,136	-0.01
Total	¥ 890,267,029	¥ 795,469,256	¥ 94,797,773	0.12

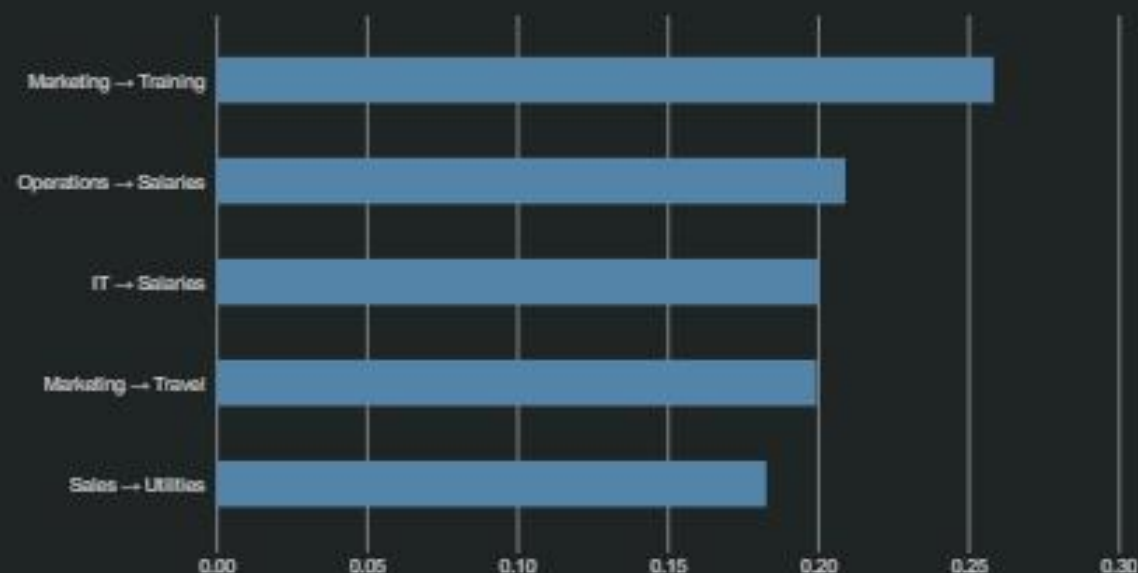
Department Category Overview

Department	Infrastructure	Marketing	Salaries	Training	Travel	Unknown	Utilities	Total
Finance	0.10	0.10	0.14	0.06	0.04		0.08	0.08
HR	0.15	0.10	0.14	0.16	0.13		0.11	0.13
IT	0.08	0.10	0.20	0.02	0.11		0.15	0.11
Marketing	0.07	0.15	0.17	0.26	0.20		0.07	0.15
Operations	0.08	0.18	0.21	0.00	0.08	-0.62	0.17	0.11
Sales	0.13	0.12	0.12	0.08	0.10		0.18	0.12
Unknown	-0.01							-0.01
Total	0.10	0.13	0.16	0.10	0.11	-0.62	0.13	0.12

Overspend % by Department



Top 5 Overspend by Department Category



4. Salaries were the largest expense-category driver

Salary	-	₦20.61M 16.21%
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This was followed by:

Utilities	-	₦17.37M
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Marketing	-	₦16.30M
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Travel	-	₦14.92M
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All major expense categories exceeded their allocated budgets.

5. East recorded the largest regional overspend

Regional analysis showed:

East: ₦21.39M | 13.40%

East had the largest regional overspend, while North recorded the smallest:

North: ₦16.53M | 10.47%

Since every region exceeded budget, the issue appears to be geographically widespread rather than isolated to one location.

Regional Performance

Executive Dashboard

Time & Budget Perf.

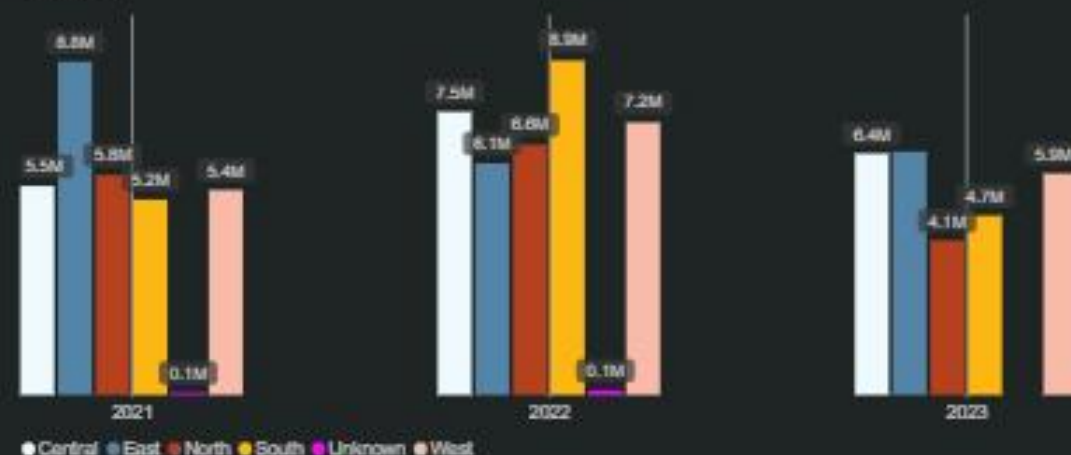
Department & Expense...

Regional Performance & Management Actions

Glossary

Region	Actual Spending	Total Budget	BV	BV %
Central	¥182,078,152	¥162,657,334.00	19420818	0.12
East	¥181,076,974	¥159,685,797.00	21391177	0.13
South	¥176,760,699	¥157,991,439.00	18769260	0.12
West	¥175,505,843	¥157,004,504.00	18501339	0.12
North	¥174,461,702	¥157,929,548.00	16532154	0.10
Unknown	¥383,659	¥200,634.00	183025	0.91

Yearly Regional Overspend
Variance %



Department Category

Actual Spendings

Department Category	Actual Spendings
Fin...	Intra...
HR	Mark...
IT	Sale...
Mark...	Train...
Oper...	Travel
Sales	Unk...
Unk...	Utilities

Central	100%
East	N 182M
South	N 181M
West	N 177M
North	N 176M
Unknown	N 174M
	N 0M
	0.2%

Total Regional Overspend



Recommendations

01 Review salary expenditure

Salaries were the largest expense-category driver. Management should review staffing levels, overtime, salary adjustments and the assumptions used to establish salary budgets.

02 Strengthen Marketing spending controls

Marketing recorded the largest departmental overspend, with Training and Travel among its largest department-category gaps. These areas should be reviewed for spending efficiency and budget adherence.

Because actual spending exceeded budget consistently, management should introduce regular budget-vs-actual reviews and establish variance thresholds that trigger investigation before overruns become significant.

03 Introduce regular budget monitoring

04 Investigate East Region

East recorded the largest regional overspend. Management should drill further into its departmental and category-level spending to determine the underlying causes.

Overspending occurred across departments, categories and regions. This suggests that management should not only focus on controlling expenditure but also assess whether budget allocations are realistic and aligned with actual operating requirements.

05 Reassess the Budgeting process

Business Value

The analysis transforms raw financial transactions into a structured view of where, when and why budget overruns are occurring.

Rather than focusing only on total spending, the analysis provides management with multiple levels of visibility, from overall performance to specific department-category combinations and regional drivers.

This supports more targeted budget monitoring, cost control and future budget planning.

Conclusion

The company exceeded its allocated budget by ~~£~~94.8M across the three-year period. Overspending was widespread, with Marketing, Salaries and the East region emerging as key areas requiring attention.

The analysis indicates that improving budget performance will require both stronger expenditure monitoring and a review of whether current budget allocations accurately reflect operational spending requirements.